



1. Description

1.1. Project

Project Name	pcm_v0_1
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	11/21/2025

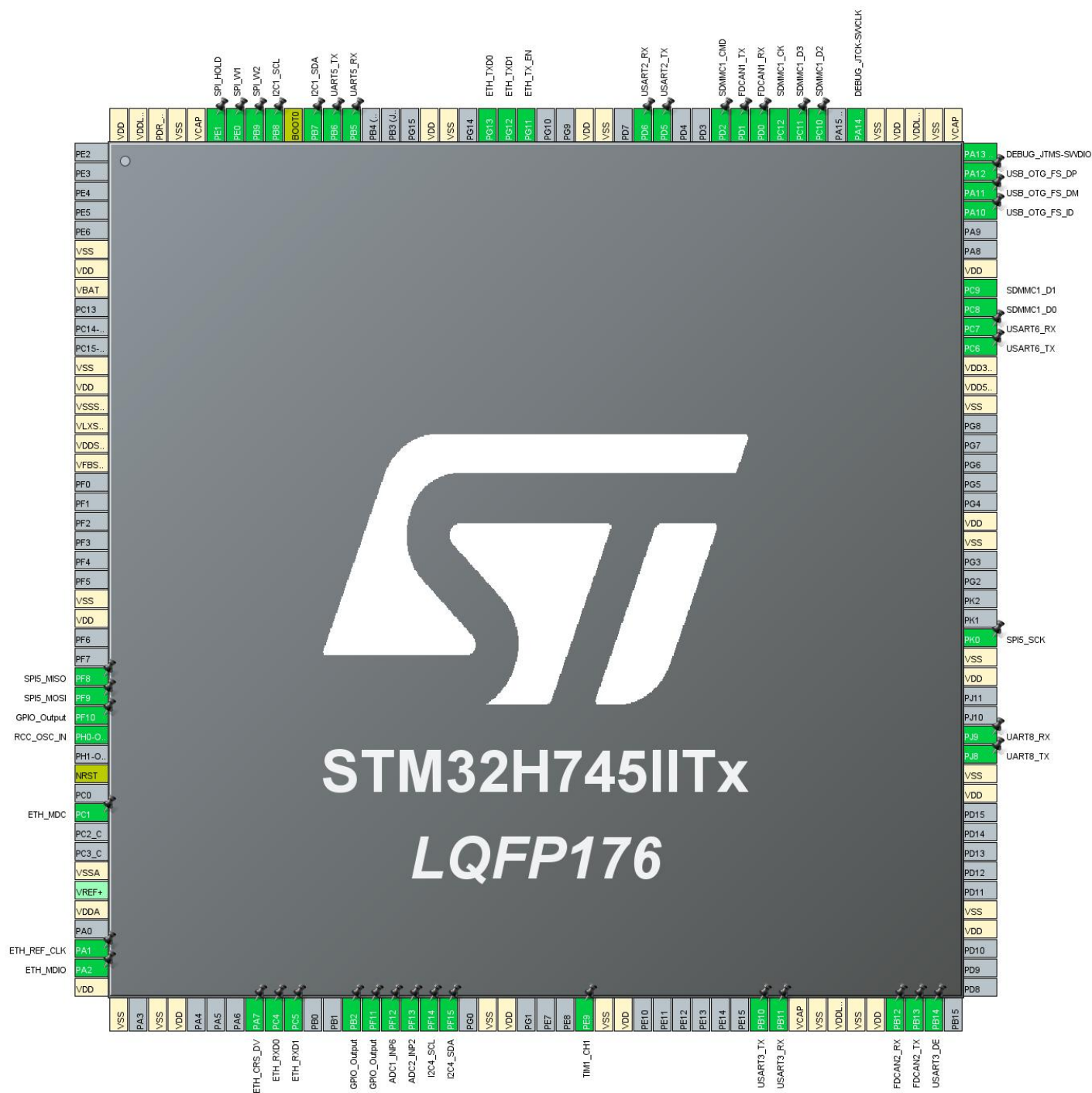
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H745/755
MCU name	STM32H745IITx
MCU Package	LQFP176
MCU Pin number	176

1.3. Core(s) information

Core(s)	ARM Cortex-M7 ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

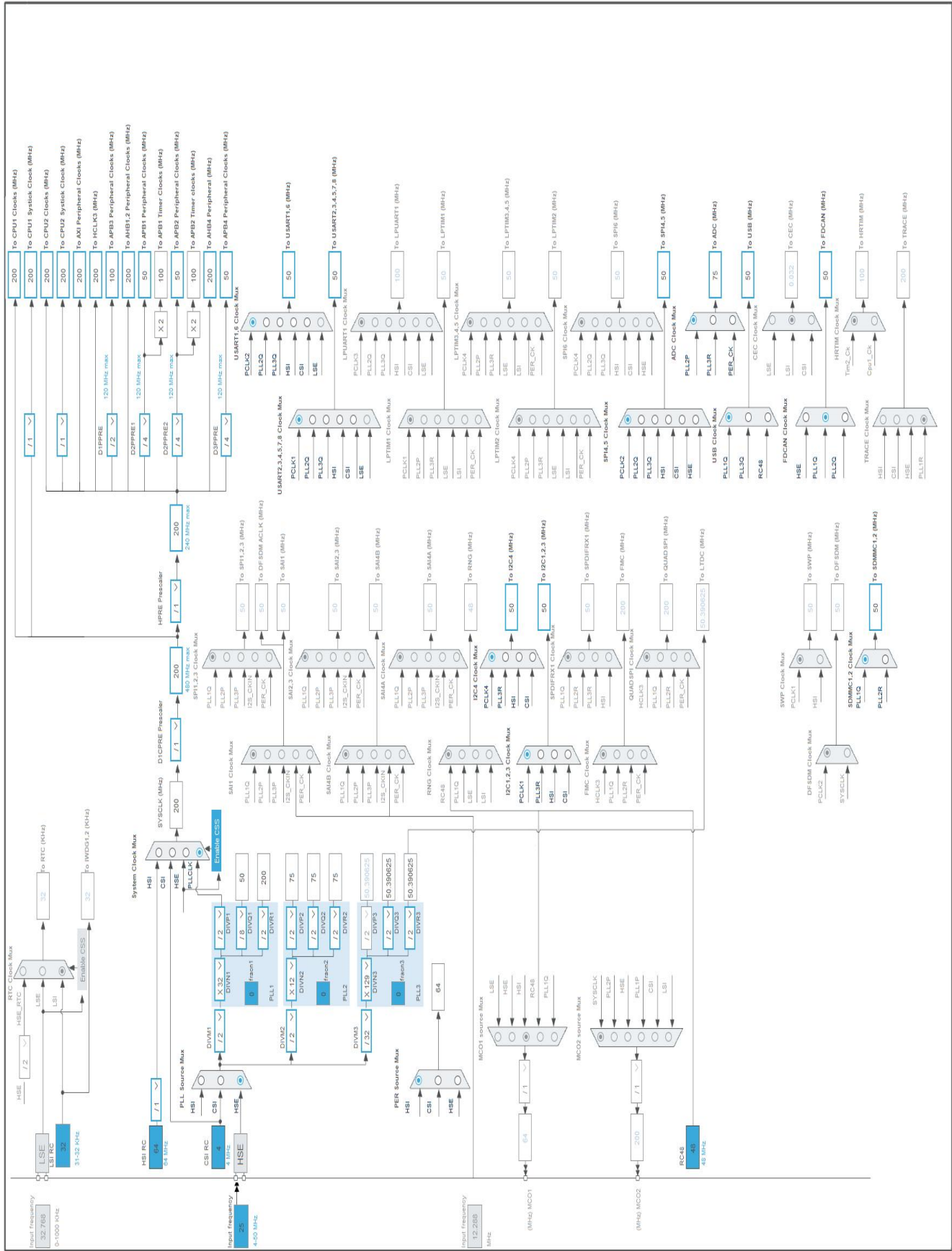
Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
6	VSS	Power		
7	VDD	Power		
8	VBAT	Power		
12	VSS	Power		
13	VDD	Power		
14	VSSSMPS	Power		
15	VLXSMPS	Power		
16	VDDSMPS	Power		
17	VFBSMPS	Power		
24	VSS	Power		
25	VDD	Power		
28	PF8	I/O	SPI5_MISO	
29	PF9	I/O	SPI5_MOSI	
30	PF10 *	I/O	GPIO_Output	
31	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
33	NRST	Reset		
35	PC1	I/O	ETH_MDC	
38	VSSA	Power		
40	VDDA	Power		
42	PA1	I/O	ETH_REF_CLK	
43	PA2	I/O	ETH_MDIO	
44	VDD	Power		
45	VSS	Power		
47	VSS	Power		
48	VDD	Power		
52	PA7	I/O	ETH_CRS_DV	
53	PC4	I/O	ETH_RXD0	
54	PC5	I/O	ETH_RXD1	
57	PB2 *	I/O	GPIO_Output	
58	PF11 *	I/O	GPIO_Output	
59	PF12	I/O	ADC1_INP6	
60	PF13	I/O	ADC2_INP2	
61	PF14	I/O	I2C4_SCL	
62	PF15	I/O	I2C4_SDA	
64	VSS	Power		
65	VDD	Power		

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
69	PE9	I/O	TIM1_CH1	
70	VSS	Power		
71	VDD	Power		
78	PB10	I/O	USART3_TX	
79	PB11	I/O	USART3_RX	
80	VCAP	Power		
81	VSS	Power		
82	VDDLDO	Power		
83	VSS	Power		
84	VDD	Power		
85	PB12	I/O	FDCAN2_RX	
86	PB13	I/O	FDCAN2_TX	
87	PB14	I/O	USART3_DE	
92	VDD	Power		
93	VSS	Power		
99	VDD	Power		
100	VSS	Power		
101	PJ8	I/O	UART8_TX	
102	PJ9	I/O	UART8_RX	
105	VDD	Power		
106	VSS	Power		
107	PK0	I/O	SPI5_SCK	
112	VSS	Power		
113	VDD	Power		
119	VSS	Power		
120	VDD50_USB	Power		
121	VDD33_USB	Power		
122	PC6	I/O	USART6_TX	
123	PC7	I/O	USART6_RX	
124	PC8	I/O	SDMMC1_D0	
125	PC9	I/O	SDMMC1_D1	
126	VDD	Power		
129	PA10	I/O	USB_OTG_FS_ID	
130	PA11	I/O	USB_OTG_FS_DM	
131	PA12	I/O	USB_OTG_FS_DP	
132	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
133	VCAP	Power		
134	VSS	Power		
135	VDDLDO	Power		

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
136	VDD	Power		
137	VSS	Power		
138	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
140	PC10	I/O	SDMMC1_D2	
141	PC11	I/O	SDMMC1_D3	
142	PC12	I/O	SDMMC1_CK	
143	PD0	I/O	FDCAN1_RX	
144	PD1	I/O	FDCAN1_TX	
145	PD2	I/O	SDMMC1_CMD	
148	PD5	I/O	USART2_TX	
149	PD6	I/O	USART2_RX	
151	VSS	Power		
152	VDD	Power		
155	PG11	I/O	ETH_TX_EN	
156	PG12	I/O	ETH_TXD1	
157	PG13	I/O	ETH_TXD0	
159	VSS	Power		
160	VDD	Power		
164	PB5	I/O	UART5_RX	
165	PB6	I/O	UART5_TX	
166	PB7	I/O	I2C1_SDA	
167	BOOT0	Boot		
168	PB8	I/O	I2C1_SCL	
169	PB9 *	I/O	GPIO_Output	SPI_W2
170	PE0 *	I/O	GPIO_Output	SPI_W1
171	PE1 *	I/O	GPIO_Output	SPI_HOLD
172	VCAP	Power		
173	VSS	Power		
174	PDR_ON	Power		
175	VDDLDO	Power		
176	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H745/755
MCU	STM32H745IITx
Datasheet	DS12923 Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

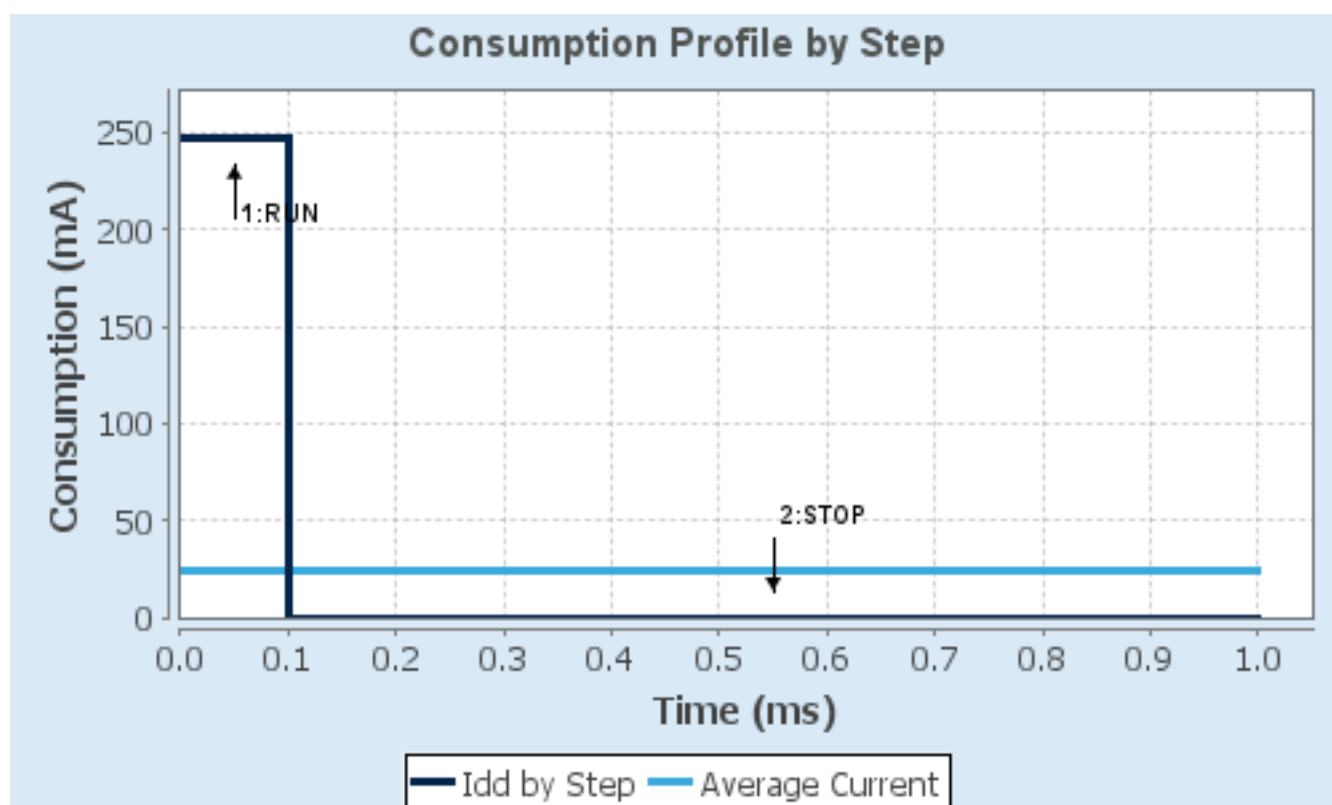
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN/CRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	CM7: ITCM/Cache / CM4: FLASH_B/ART	CM7: NA / CM4: NA
CM7 Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL ALL_IPs_ON	LSE Flash-ON
CM4 Frequency	240 MHz	0 Hz
Clock Source Frequency	25 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	247 mA	145 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	24.83 mA
Battery Life	1 month, 29 days, 21 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	pcm_v0_1
Project Folder	D:\hashtic\ST_workspaces\HASHTIC\pcm_v0_1
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	M4-0x200
Minimum Stack Size	M4-0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls ARM Cortex-M7

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO

2.4. Advanced Settings - Generated Function Calls ARM Cortex-M4

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_ADC1_Init	ADC1
3	MX_ADC2_Init	ADC2
4	MX_ETH_Init	ETH

Rank	Function Name	Peripheral Instance Name
5	MX_FDCAN1_Init	FDCAN1
6	MX_FDCAN2_Init	FDCAN2
7	MX_I2C1_Init	I2C1
8	MX_I2C4_Init	I2C4
9	MX_SDMMC1_SD_Init	SDMMC1
10	MX_SPI5_Init	SPI5
11	MX_TIM1_Init	TIM1
12	MX_UART5_Init	UART5
13	MX_USART2_UART_Init	USART2
14	MX_USART3_UART_Init	USART3
15	MX_USART6_UART_Init	USART6
16	MX_USB_OTG_FS_USB_Init	USB_OTG_FS
17	MX_FREERTOS_Init	FREERTOS_M4
18	MX_UART8_Init	UART8

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN6

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

ADCs_Common_Settings:

Mode	Independent mode
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ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 4 *
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Oversampling Ratio	1
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 6
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.2. ADC2

IN2: IN2 Single-ended

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M4

Initialized Context: Cortex-M4

Power Domain: D2

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler **Asynchronous clock mode divided by 4 ***

Resolution ADC 16-bit resolution

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Overrun behaviour Overrun data preserved

Left Bit Shift No bit shift

Conversion Data Management Mode Regular Conversion data stored in DR register only

Low Power Auto Wait Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Oversampling Ratio 1

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 2

Sampling Time 1.5 Cycles

Offset Number No offset

Offset Signed Saturation Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.3. CORTEX_M7

3.3.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

Speculation default mode Settings:

Speculation default mode **Enabled ***

Cortex Interface Settings:

CPU ICache Disabled

CPU DCache Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region Enabled

MPU Region Base Address **0x0 ***

MPU Region Size 4GB

MPU SubRegion Disable **0x87 ***

MPU TEX field level level 0

MPU Access Permission ALL ACCESS NOT PERMITTED

MPU Instruction Access DISABLE

MPU Shareability Permission ENABLE

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.4. DEBUG

Debug: Serial Wire

3.4.1. Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	

3.5. ETH

Mode: RMII

3.5.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

General : Ethernet Configuration:

Warning	Ethernet RX and Tx descriptors needs to be placed in a RAM memory accessible by the dedicated Ethernet DMA
Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
First Tx Descriptor Address	0x30000080 *
Rx Descriptor Length	4
First Rx Descriptor Address	0x30000000 *
Rx Buffers Address	0x30000100 *
Rx Buffers Length	1536

3.6. FDCAN1

mode: Activated

3.6.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1

Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	0
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable
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Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	320.0 *
Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	960 *
Nominal Baud Rate	1041666 *

3.7. FDCAN2

mode: Activated

3.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Frame Format	Classic mode
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Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	0
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable
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Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	320.0 *
Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	960 *
Nominal Baud Rate	1041666 *

3.8. I2C1

I2C: I2C

3.8.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
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Initialized Context:	Cortex-M4
Power Domain:	D2
Timing configuration:	
Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00C0EAFB *
Slave Features:	
Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.9. I2C4

I2C: I2C

3.9.1. Parameter Settings:

Core(s) Settings:	
Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D3
Timing configuration:	
Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00C0EAFB *
Slave Features:	
Clock No Stretch Mode	Disabled

General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.10. MEMORYMAP

mode: Activated

3.10.1. Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

3.11. RCC

High Speed Clock (HSE): BYPASS Clock Source

3.11.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D3

Power Parameters:

SupplySource	PWR_DIRECT_SMPS_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 3

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	MEDIUM VCO range

3.12. SDMMC1

Mode: SD 4 bits Wide bus

3.12.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D1

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.13. SPI5

Mode: Full-Duplex Master

3.13.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
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Baud Rate	25.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.14. SYS_M4

Timebase Source: SysTick

3.14.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	

3.15. SYS

Timebase Source: SysTick

3.15.1. Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

3.16. TIM1

Channel1: PWM Generation CH1

3.16.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable

Lock Configuration	Off
Clear Input:	
Clear Input Source	Disable
PWM Generation Channel 1:	
Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.17. UART5

Mode: Asynchronous

3.17.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable

Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.18. UART8

Mode: Asynchronous

3.18.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.19. USART2

Mode: Asynchronous

3.19.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.20. USART3

Mode: Asynchronous

mode: Hardware Flow Control (RS485)

3.20.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
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Initialized Context:	Cortex-M4
Power Domain:	D2
Basic Parameters:	
Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Polarity	High
Assertion Time	0
Deassertion Time	0
Advanced Features:	
Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.21. USART6

Mode: Asynchronous

3.21.1. Parameter Settings:

Core(s) Settings:	
Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2
Basic Parameters:	
Baud Rate	115200

Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.22. USB_OTG_FS

Mode: OTG/Dual_Role_Device

3.22.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

3.23. FREERTOS_M4

Interface: CMSIS_V2

3.23.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4

Power Domain:	D2
API:	
FreeRTOS API	CMSIS v2
Versions:	
FreeRTOS version	10.3.1
CMSIS-RTOS version	2.00
MPU/FPU:	
ENABLE_MPU	Disabled
ENABLE_FPU	Disabled
Kernel settings:	
USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemD2Clock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
Memory management settings:	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	15360
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled
Run time and task stats gathering related definitions:	
GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled

USE_STATS_FORMATTING_FUNCTIONS

Disabled

Co-routine related definitions:

USE_CO_ROUTINES

Disabled

MAX_CO_ROUTINE_PRIORITIES

2

Software timer definitions:

USE_TIMERS

Enabled

TIMER_TASK_PRIORITY

2

TIMER_QUEUE_LENGTH

10

TIMER_TASK_STACK_DEPTH

256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY

15

LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY

5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE

size_t

USE_POSIX_ERRNO

Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME

Enabled

USE_OS2_THREAD_ENUMERATE

Enabled

USE_OS2_EVENTFLAGS_FROM_ISR

Enabled

USE_OS2_THREAD_FLAGS

Enabled

USE_OS2_TIMER

Enabled

USE_OS2_MUTEX

Enabled

3.23.2. Include parameters:

Core(s) Settings:

Context(s):

Cortex-M4

Initialized Context:

Cortex-M4

Power Domain:

D2

Include definitions:

vTaskPrioritySet

Enabled

uxTaskPriorityGet

Enabled

vTaskDelete

Enabled

vTaskCleanUpResources

Disabled

vTaskSuspend

Enabled

vTaskDelayUntil

Enabled

vTaskDelay

Enabled

xTaskGetSchedulerState

Enabled

xTaskResumeFromISR

Enabled

xQueueGetMutexHolder

Enabled

xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.23.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
----------------------	----------

Project settings (see parameter description first):

Use FW pack heap file	Enabled
-----------------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
ADC1	PF12	ADC1_INP6	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
ADC2	PF13	ADC2_INP2	Analog mode	No pull-up and no pull-down	n/a		Cortex-M4	D2
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
ETH	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA7	ETH_CRS_DV	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PG11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PG12	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PG13	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
FDCAN1	PD0	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PD1	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
FDCAN2	PB12	FDCAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB13	FDCAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
I2C1	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
I2C4	PF14	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D3
	PF15	I2C4_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D3
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
SDMMC1	PC8	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M4	D1
SPI5	PF8	SPI5_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PF9	SPI5_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PK0	SPI5_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
UART5	PB5	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB6	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
UART8	PJ8	UART8_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PJ9	UART8_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
USART2	PD5	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PD6	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB14	USART3_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
USART6	PC6	USART6_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PC7	USART6_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
USB_OTG_FS	PA10	USB_OTG_FS_ID	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
GPIO	PF10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PF11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI_W2	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI_W1	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI_HOLD	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

* Initialized context

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CM4 send event interrupt for CM7	unused		
FPU global interrupt	unused		
HSEM1 global interrupt	unused		
Hold core interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0

Interrupt Table	Enable	Preenmption Priority	SubPriority
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
PVD and AVD interrupts through EXTI line 16		unused	
Flash global interrupt		unused	
ADC1 and ADC2 global interrupts		unused	
FDCAN1 interrupt 0		unused	
FDCAN2 interrupt 0		unused	
FDCAN1 interrupt 1		unused	
FDCAN2 interrupt 1		unused	
TIM1 break interrupt		unused	
TIM1 update interrupt		unused	
TIM1 trigger and commutation interrupts		unused	
TIM1 capture compare interrupt		unused	
I2C1 event interrupt		unused	
I2C1 error interrupt		unused	
USART2 global interrupt		unused	
USART3 global interrupt		unused	
SDMMC1 global interrupt		unused	
UART5 global interrupt		unused	
Ethernet global interrupt		unused	
Ethernet wake-up interrupt through EXTI line 86		unused	
FDCAN calibration unit interrupt		unused	
CM7 send event interrupt for CM4		unused	
USART6 global interrupt		unused	
FPU global interrupt		unused	
UART8 global interrupt		unused	
SPI5 global interrupt		unused	
I2C4 event interrupt		unused	
I2C4 error interrupt		unused	
HSEM2 global interrupt		unused	
Hold core interrupt		unused	

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Category view Context Execution view Context Initialization view Power Domain view

   Choose filters ...

... by Context Execution: ☐ Cortex-M7 ☐ Cortex-M4

... by Context Initialization: ☐ Cortex-M7 ☐ Cortex-M4 ☒ None

... by Power Domain: ☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

FREERTOS_M4 ✓

System Core	Analog	Timers	Connectivity		Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
BDMA	ADC1 ✓	TIM1 ✓	ETH ✓	FDCAI1 ✓				DEBUG ✓			
CORTEX_M4	ADC2 ✓		FDCAI2 ✓	I2C1 ✓							
CORTEX_M7 ✓			I2C4 ✓	SDMMC1 ✓							
DMA			SPI5 ✓	UART5 ✓							
GPIO ✓			UART8 ✓	USART2 ✓							
MDMA			USART3 ✓	USART6 ✓							
NVIC1 ✓			USB_FS ✓								
NVIC2 ✓											
RCC ✓											
SYS_M4 ✓											
SYS_M7 ✓											

5.1.2. Without filters

[Category view](#)
[Context Execution view](#)
[Context Initialization view](#)
[Power Domain view](#)




 Choose filters ...

... by Context Execution ...
☐ Cortex-M7 ☐ Cortex-M4

... by Context Initialization ...
☐ Cortex-M7 ☐ Cortex-M4 ☒ None

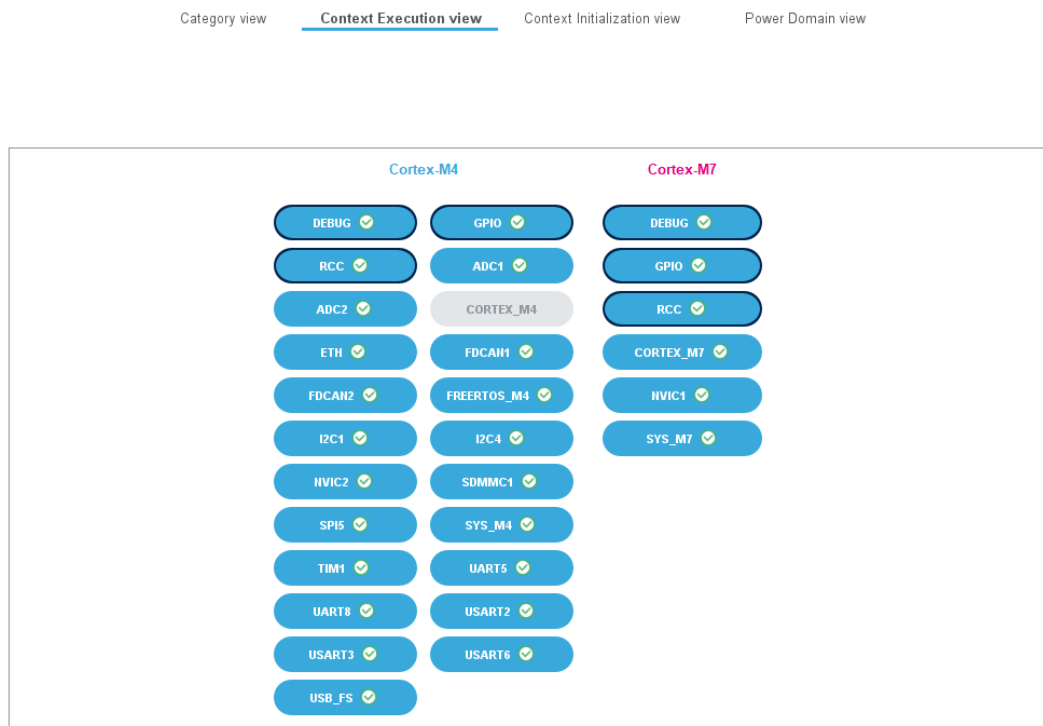
... by Power Domain ...
☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

FREERTOS_M4 ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
BDMA	ADC1 ✓	TIM1 ✓	ETH ✓	FDCAI1 ✓			DEBUG ✓			
CORTEX_M4	ADC2 ✓		FDCAI2 ✓	I2C1 ✓						
CORTEX_M7 ✓			I2C4 ✓	SDMMC1 ✓						
DMA			SPI5 ✓	UART5 ✓						
GPIO ✓			UART8 ✓	USART2 ✓						
MDMA			USART3 ✓	USART6 ✓						
NVIC1 ✓			USB_FS ✓							
NVIC2 ✓										
RCC ✓										
SYS_M4 ✓										
SYS_M7 ✓										

5.2. Context Execution view



5.3. Context Initialization view



5.4. Power Domain view



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5286-stm32h7x5x7-dualcore-microcontroller-debugging-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5557-stm32h745755-and-stm32h747757-lines-dualcore-architecture-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5617-stm32h745755-and-stm32h747757-lines-interprocessor-communications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-

stmicroelectronics.pdf

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